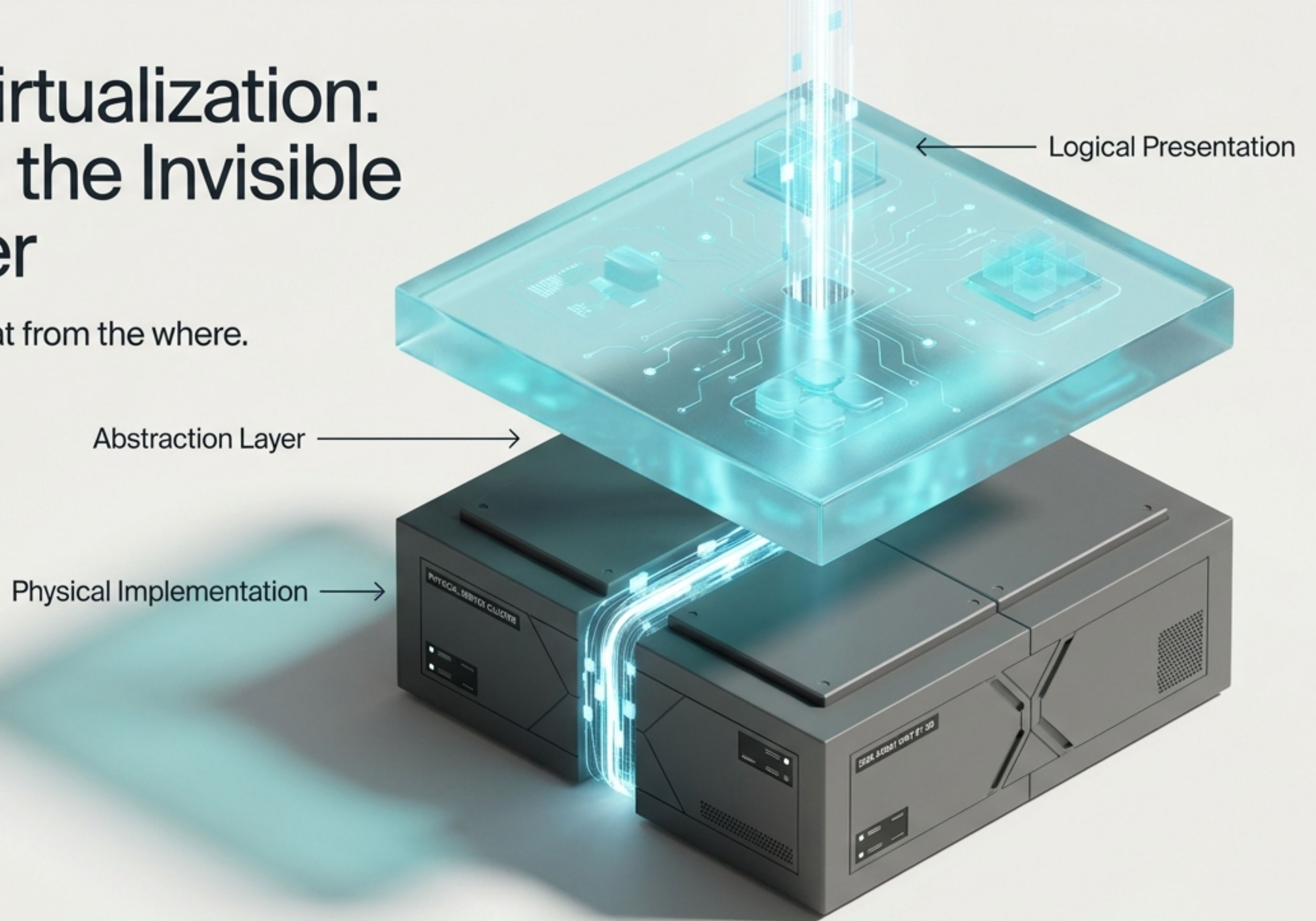


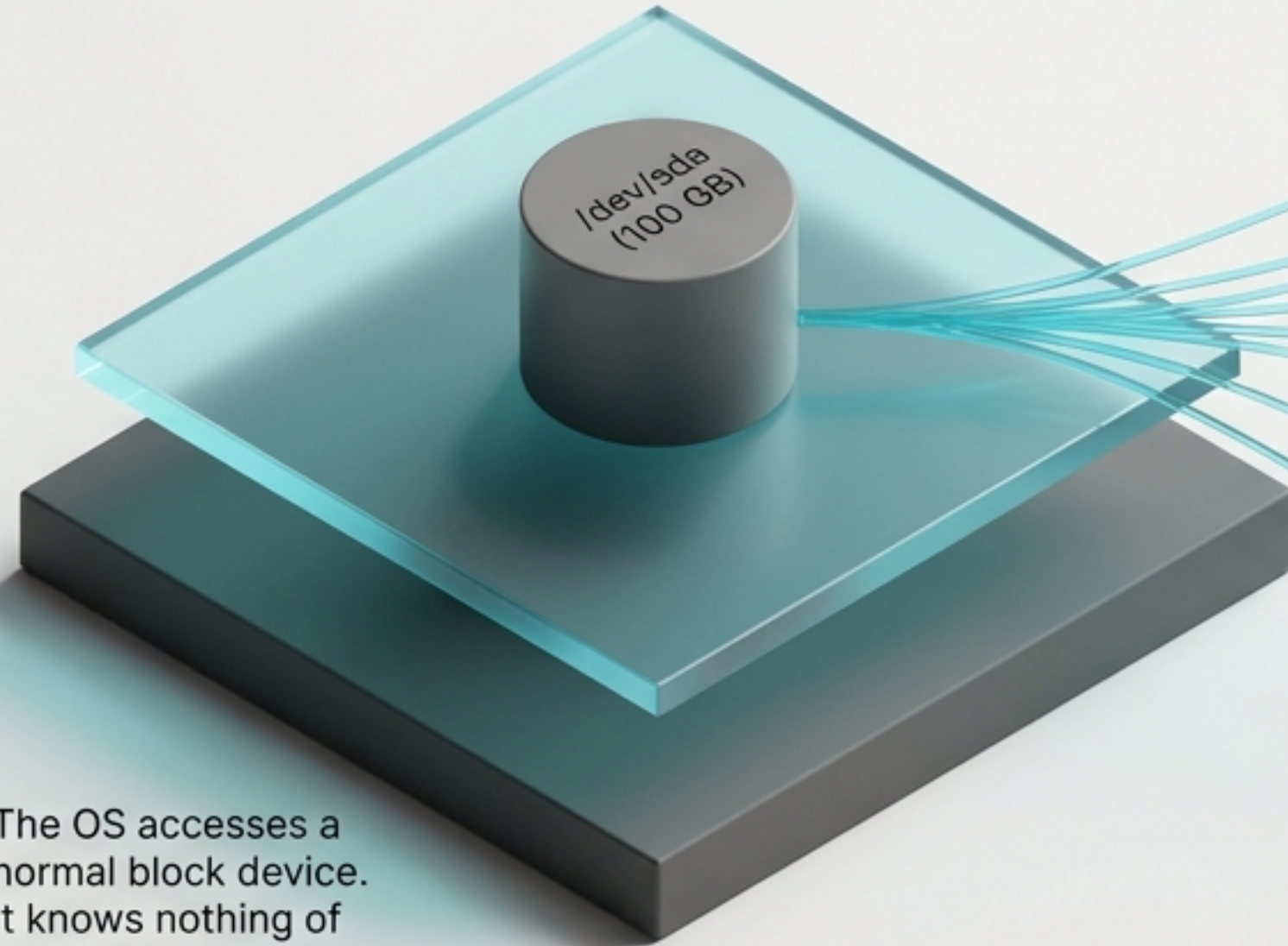
Storage Virtualization: Mastering the Invisible Datacenter

Disconnecting the what from the where.



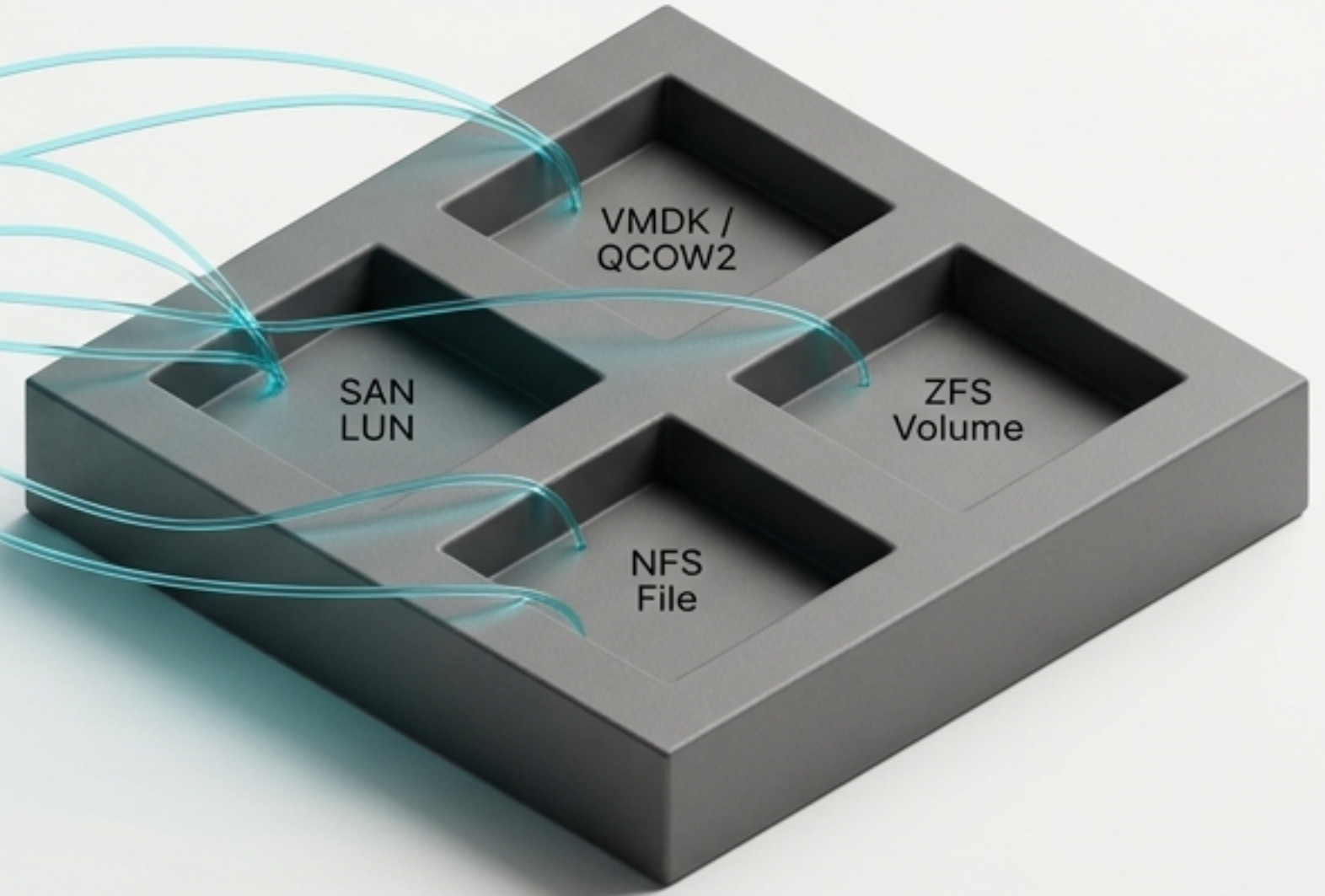
The Core Abstraction: Disconnecting the What from the Where

The Guest View



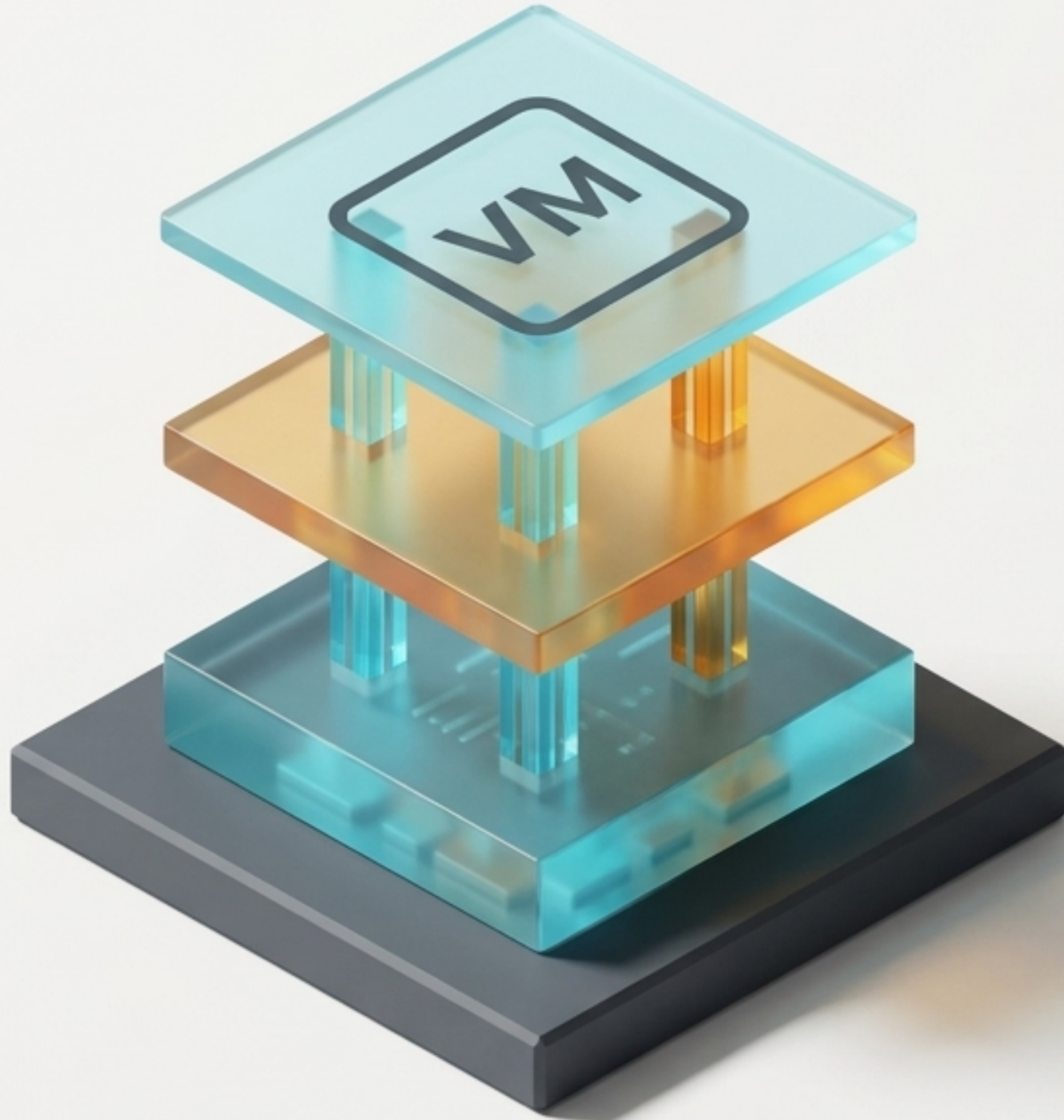
The OS accesses a normal block device. It knows nothing of the infrastructure.

The Hypervisor Reality



Storage virtualization presents logical storage resources independently of their underlying physical implementation.

The Anatomy of a Virtual Disk: Formats and Controllers

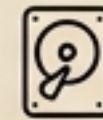


The Controller (The Translator)

Emulated (IDE, SATA, SCSI):

Maximum compatibility.

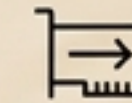
Guest uses standard hardware drivers.



Paravirtualized (VirtIO):

Maximum performance.

Guest uses hypervisor-aware drivers to reduce emulation overhead.



The Format (The Container)

VMDK
(VMware)



QCOW2
(QEMU/KVM)



VHD/VHDX
(Microsoft)



VDI
(VirtualBox)



RAW
(Direct Block)



Formats dictate features: snapshots, copy-on-write, and sparse allocation.

The Provisioning Paradox: Logical Promises vs. Physical Reality

Thick Provisioning



Explicitly reserved. Predictable allocation, lower utilization.

Thin Provisioning



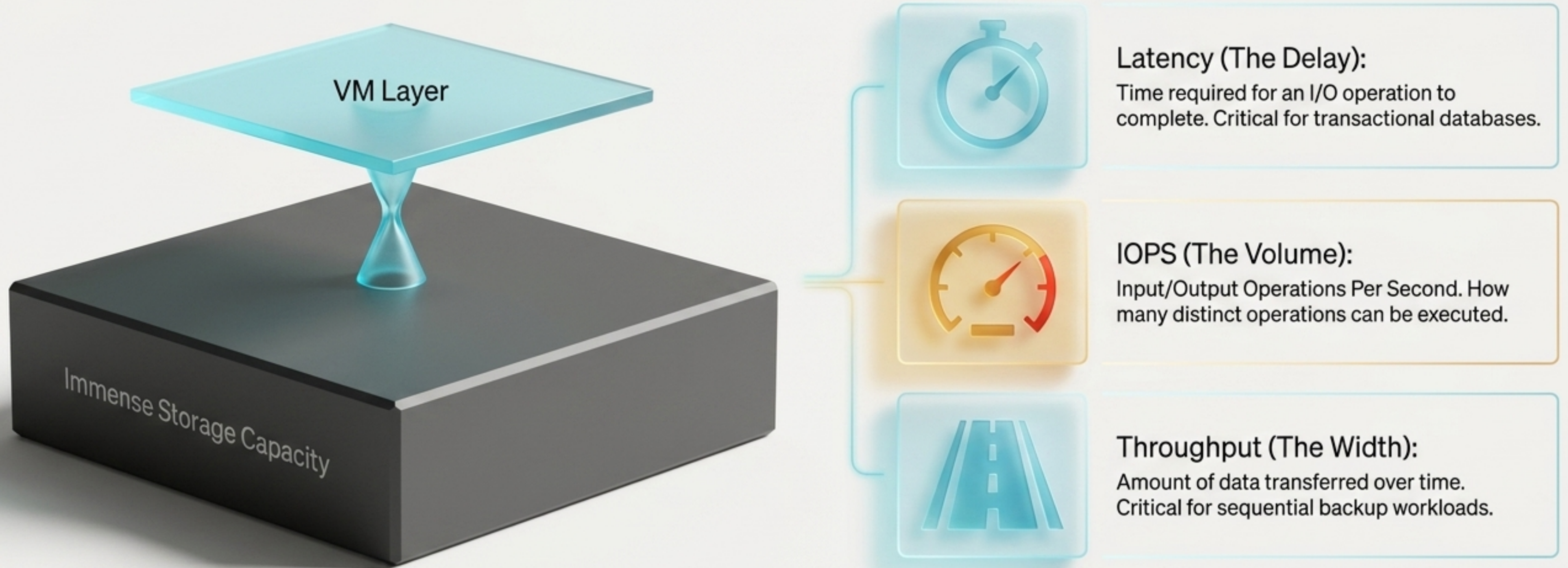
Separates logical capacity from physical capacity.
Physical storage only grows as data is written.



The Critical Risk:

Thin provisioning does not create physical space. If the physical datastore fills, the VMs **crash**—even if they believe they have logical free space.

Capacity Does Not Equal Performance



A virtual machine can have terabytes of free capacity and still grind to a halt from inadequate IOPS.

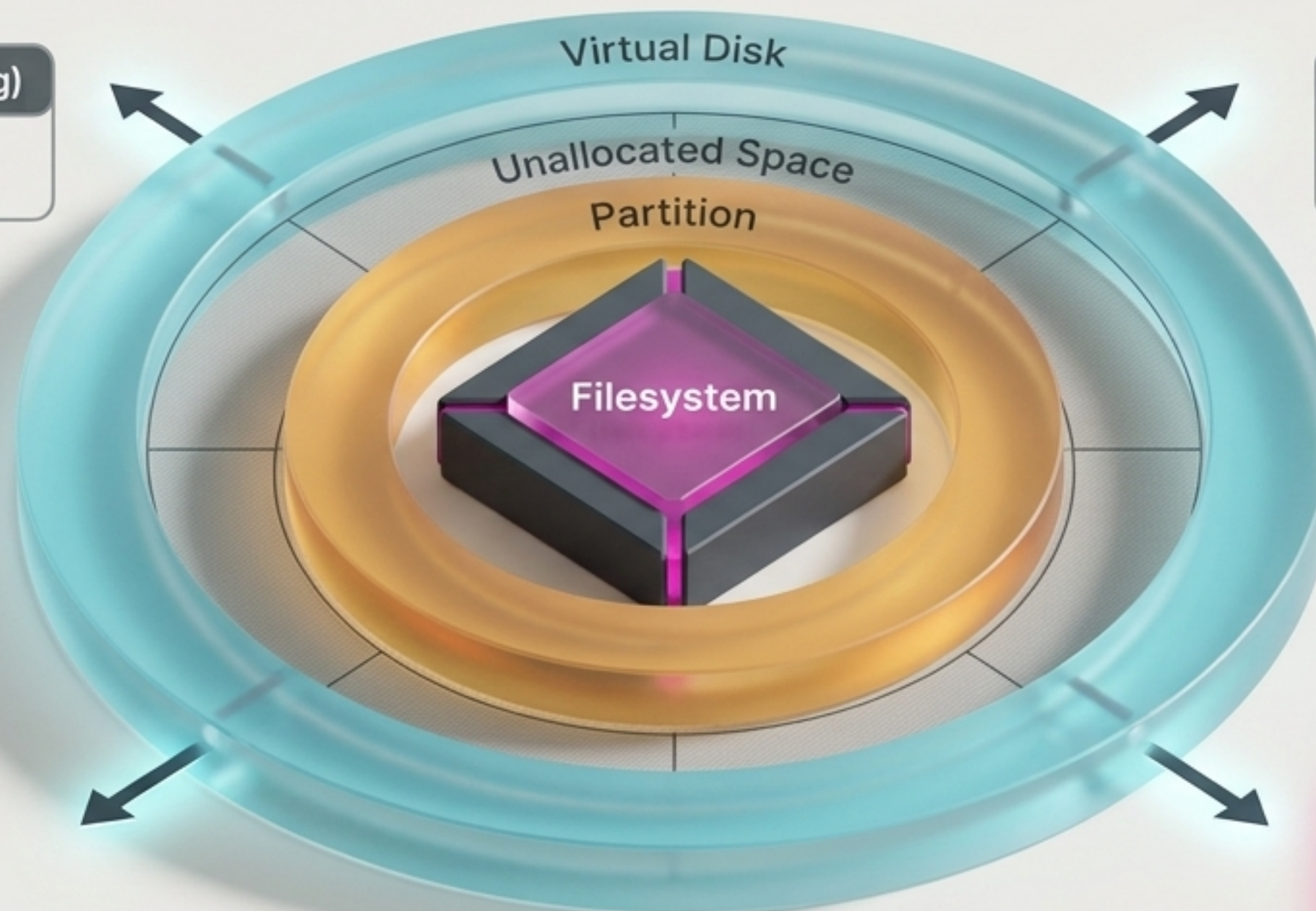
Resizing Realities: The Expansion Russian Doll

Step 1: Hypervisor Layer (Outer Ring)

Expand the virtual disk allocation. The guest sees a larger raw block device.

Step 2: Partition Layer (Middle Ring)

Expand the partition boundary to consume the newly available raw blocks.



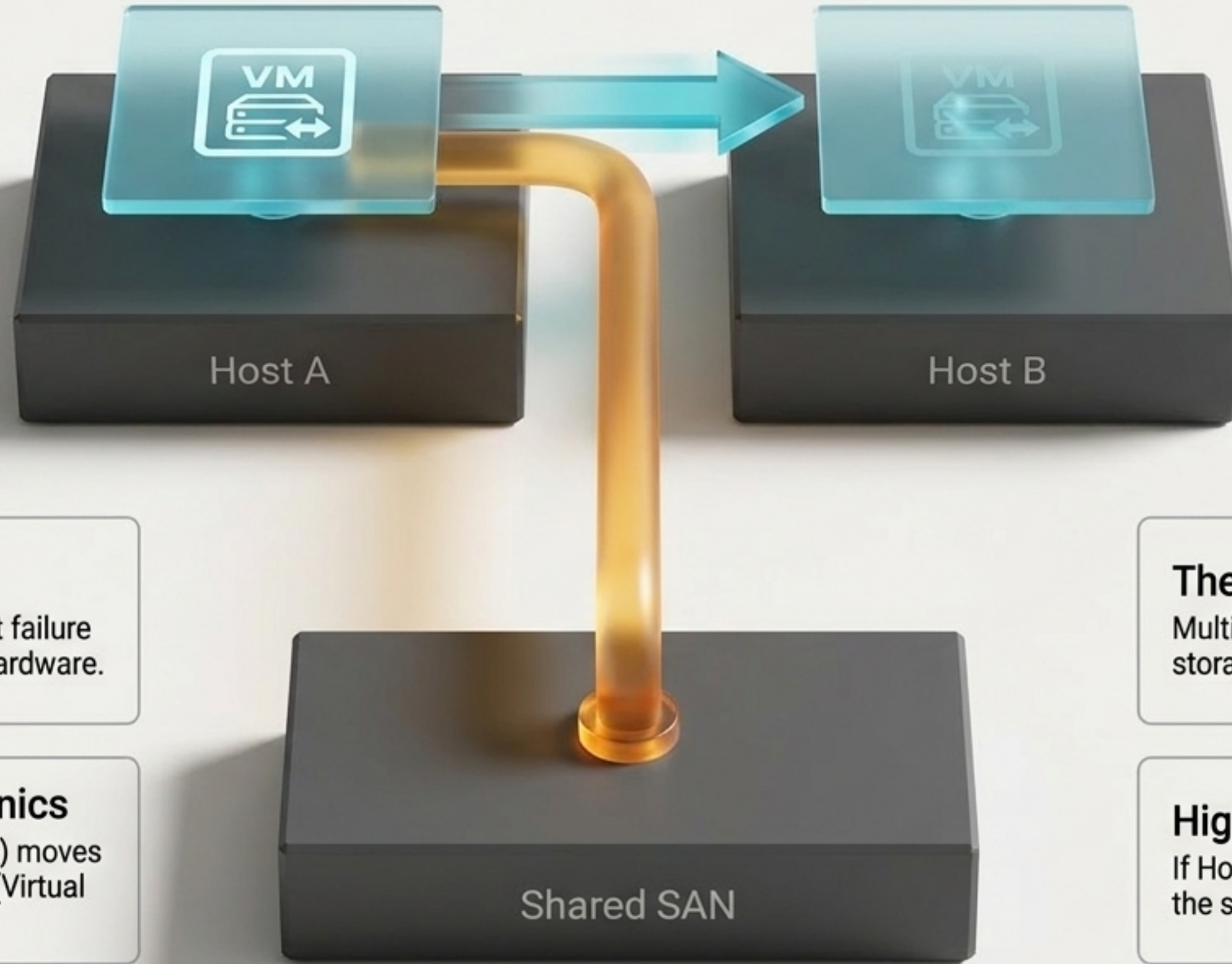
Step 3: Filesystem Layer (Inner Core)

Command the OS to index the new space.



Warning: Shrinking is highly destructive. Always proceed from the inside outward (Filesystem -> Partition -> Virtual Disk).

The Power of Shared Storage: Unlocking VM Mobility



The Local Trap

Without shared storage, a host failure traps the VM's data on dead hardware.

Live Migration Mechanics

The compute (CPU/RAM state) moves across the network. The data (Virtual Disk) stays perfectly still.

The Shared Solution

Multiple hosts access the same storage resource over the network.

High Availability

If Host A fails, Host B simply mounts the shared disk and restarts the VM.

The Shared Storage Architecture Matrix

File-Based (NAS)



Protocol: NFS

Abstraction

Storage presented as a mounted network filesystem.

Advantage

Simple management, easy file-level inspection.

Block-Based (SAN)



Protocols: iSCSI, Fibre Channel (FC)

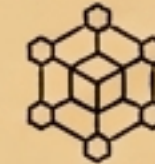
Abstraction

Storage presented as raw logical units (LUNs).

Advantage

High performance, lower overhead for database workloads.

Distributed Storage



Protocols: Ceph, vSAN

Abstraction

Storage functionality spread across multiple local host nodes.

Advantage

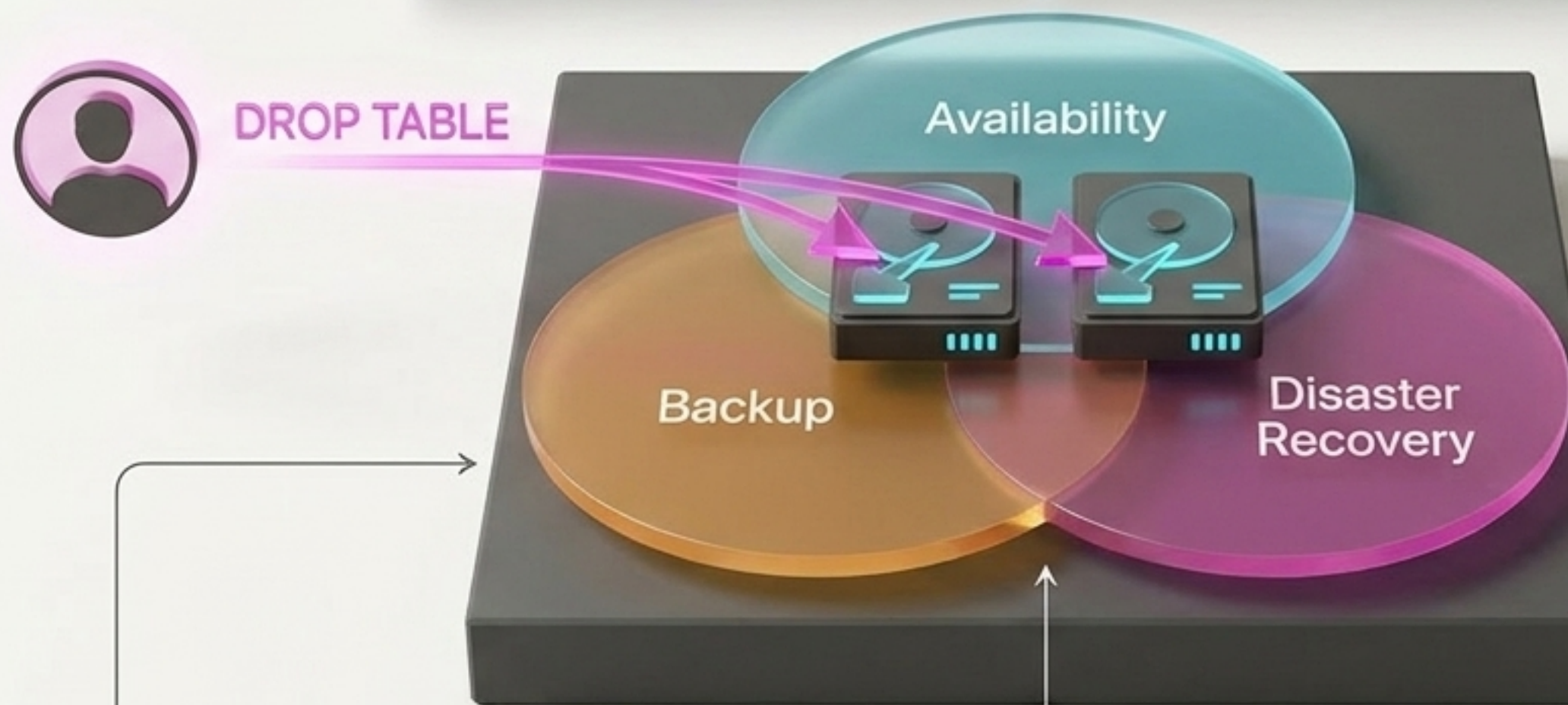
Eliminates the single central storage appliance bottleneck.

The Redundancy Spectrum: Decoding RAID

RAID Level	Technique	Fault Tolerance	Strength
RAID 0	Striping	0 Drives	Maximum Performance. No redundancy.
RAID 1	Mirroring		Ultimate Simplicity. Fast recovery.
RAID 5	Striping + Parity		Capacity Efficiency. Write penalty on rebuild.
RAID 6	Striping + Dual Parity		High Protection for large arrays.
RAID 10	Mirroring + Striping	 1+ depending on mirror	Best balance of performance and redundancy. Usable capacity is ~50%.

The Golden Rule of Virtualization: RAID is NOT Backup

The Mirroring Trap: If an administrator accidentally deletes a database, the deletion is mirrored to the redundant drive in milliseconds. RAID protects against hardware failure, not human error.



Availability (RAID / Multipathing):
Keeps services running during a hardware component failure.

Backup (Historical Copies):
Provides recoverable historical versions to protect against ransomware, corruption, or accidental deletion.

Disaster Recovery (Replication):
Restores services after a major site-level infrastructure failure.

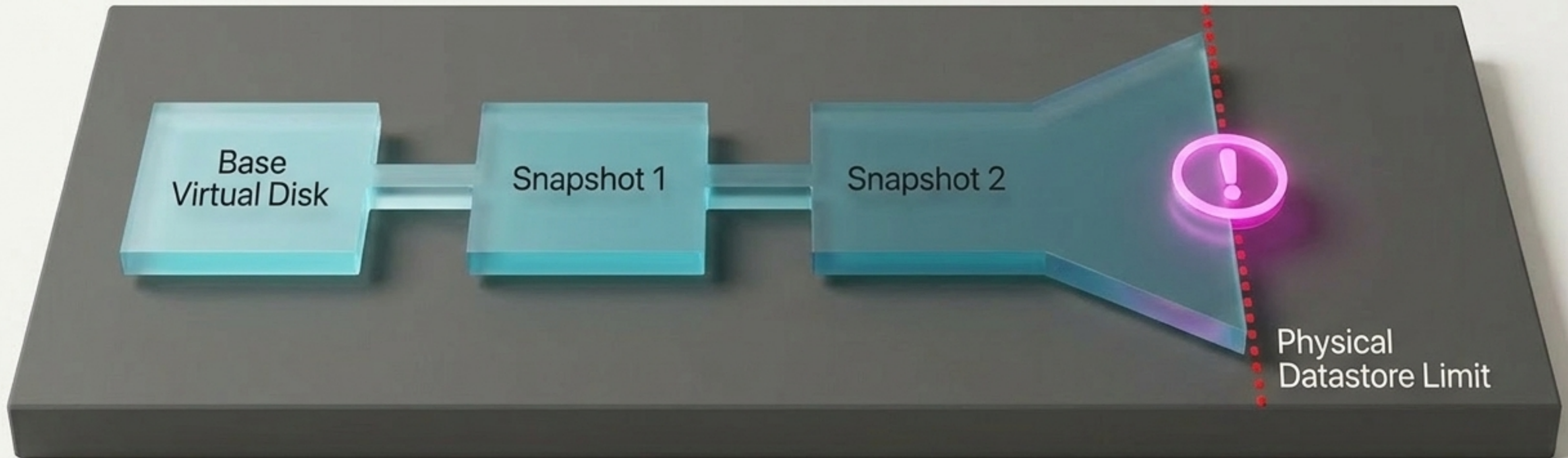
The Silent Consumers: Snapshot Lifecycle Management

The Growth Mechanic:

A snapshot captures the VM state at a point in time. As the VM writes new data, the snapshot consumes physical storage to track the changes.

Operational Mandate:

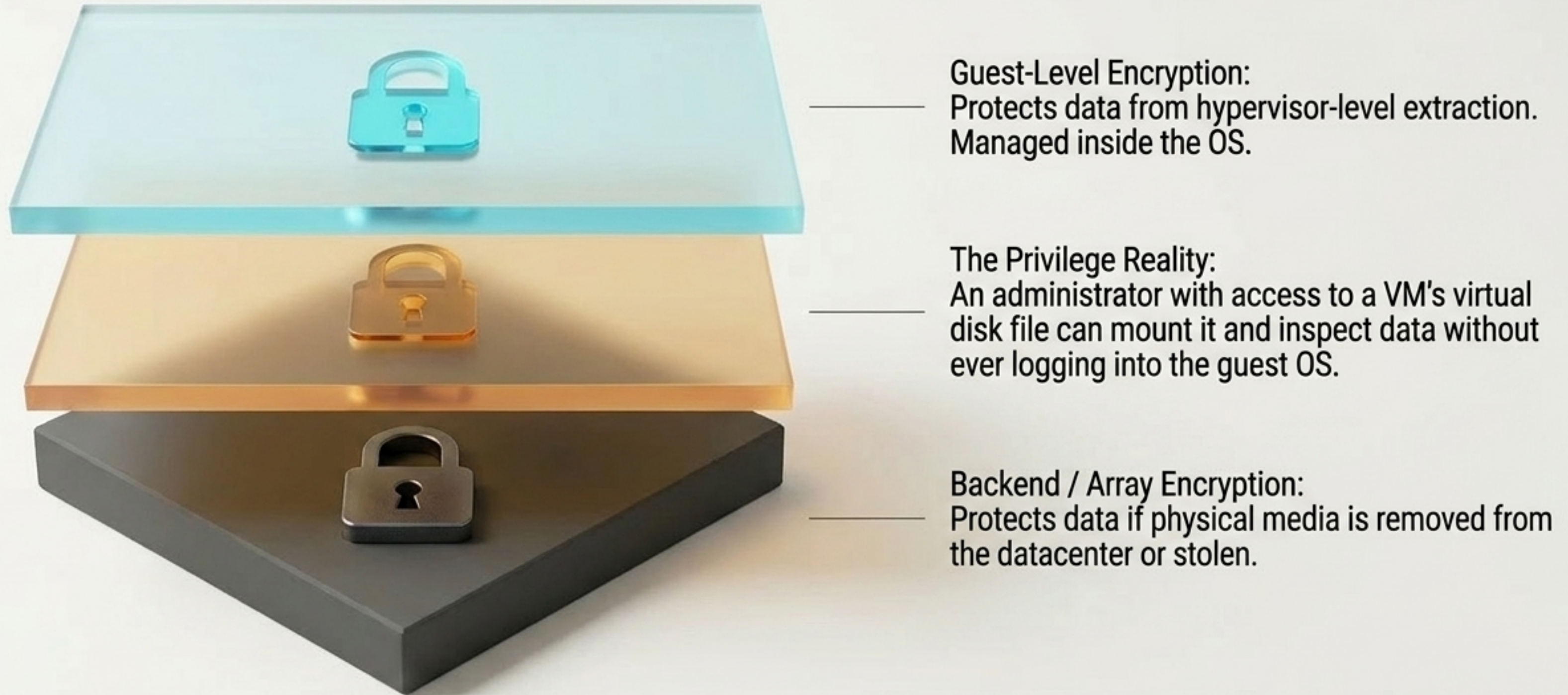
Snapshots are temporary operational tools, not backups. Every snapshot must have a documented reason, owner, and expected lifetime.



The Danger:

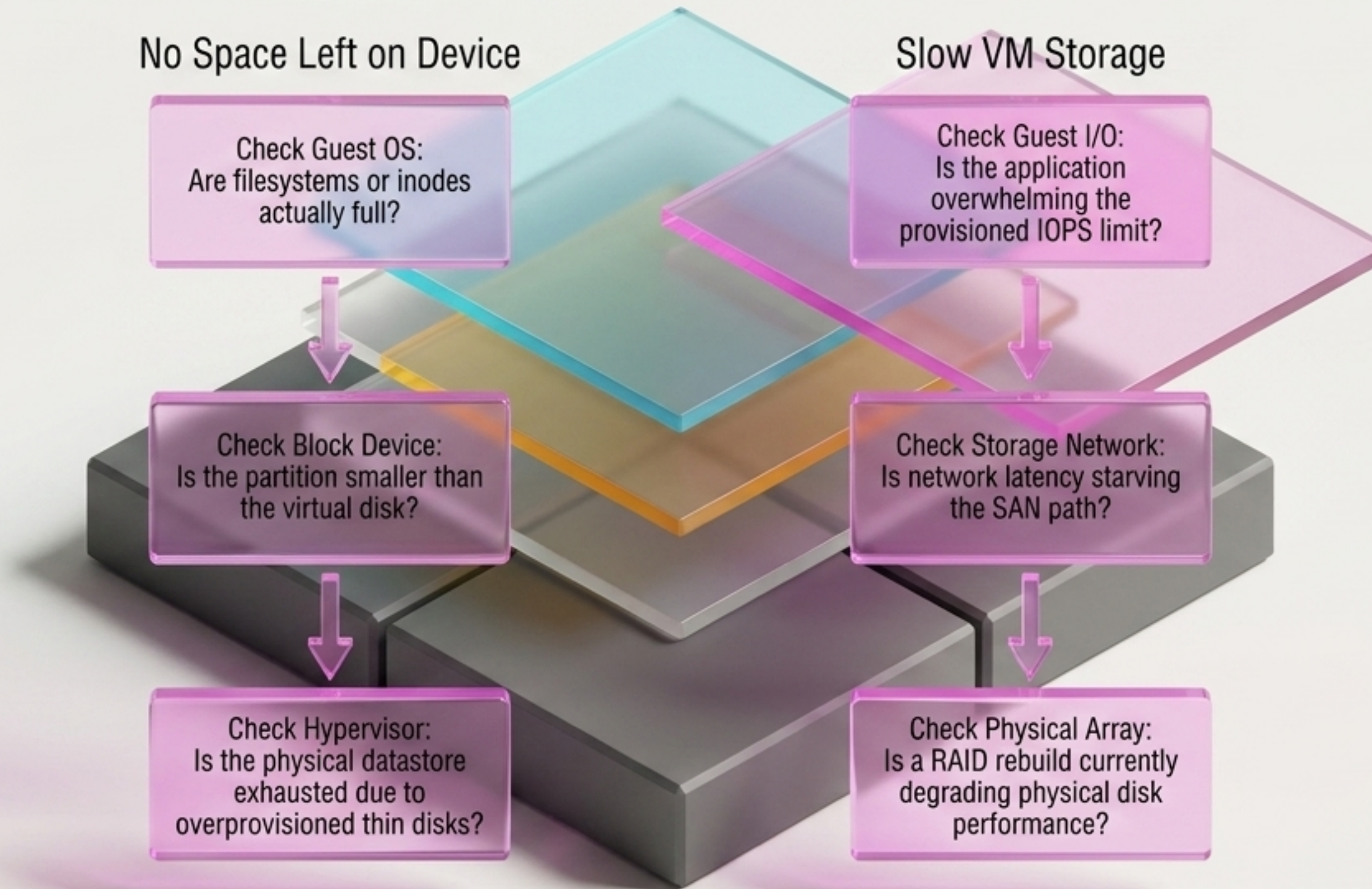
A long-lived snapshot on a heavily changing VM (like a database) will grow uncontrollably until it exhausts the datastore.

The Security Mandate: Protecting the Abstracted State



Warning: Encryption introduces key management overhead. Lost keys mean permanently unrecoverable virtual disks.

Troubleshooting the Stack: A Top-Down Diagnostic Flow



Six Pillars of Storage Architecture



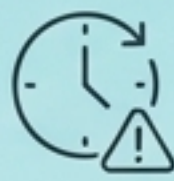
Separate Capacity from Performance:

A 10TB disk isn't automatically a fast disk. Design for IOPS first.



Understand the Full Stack:

Map guest storage all the way down to the physical backend.



Monitor Thin Provisioning:

Treat logical capacity that exceeds physical reality as a ticking clock.



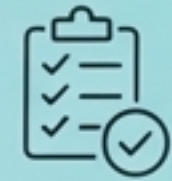
RAID != Backup:

Build distinct strategies for Availability, Recovery, and Disaster.



Design for Failure Domains:

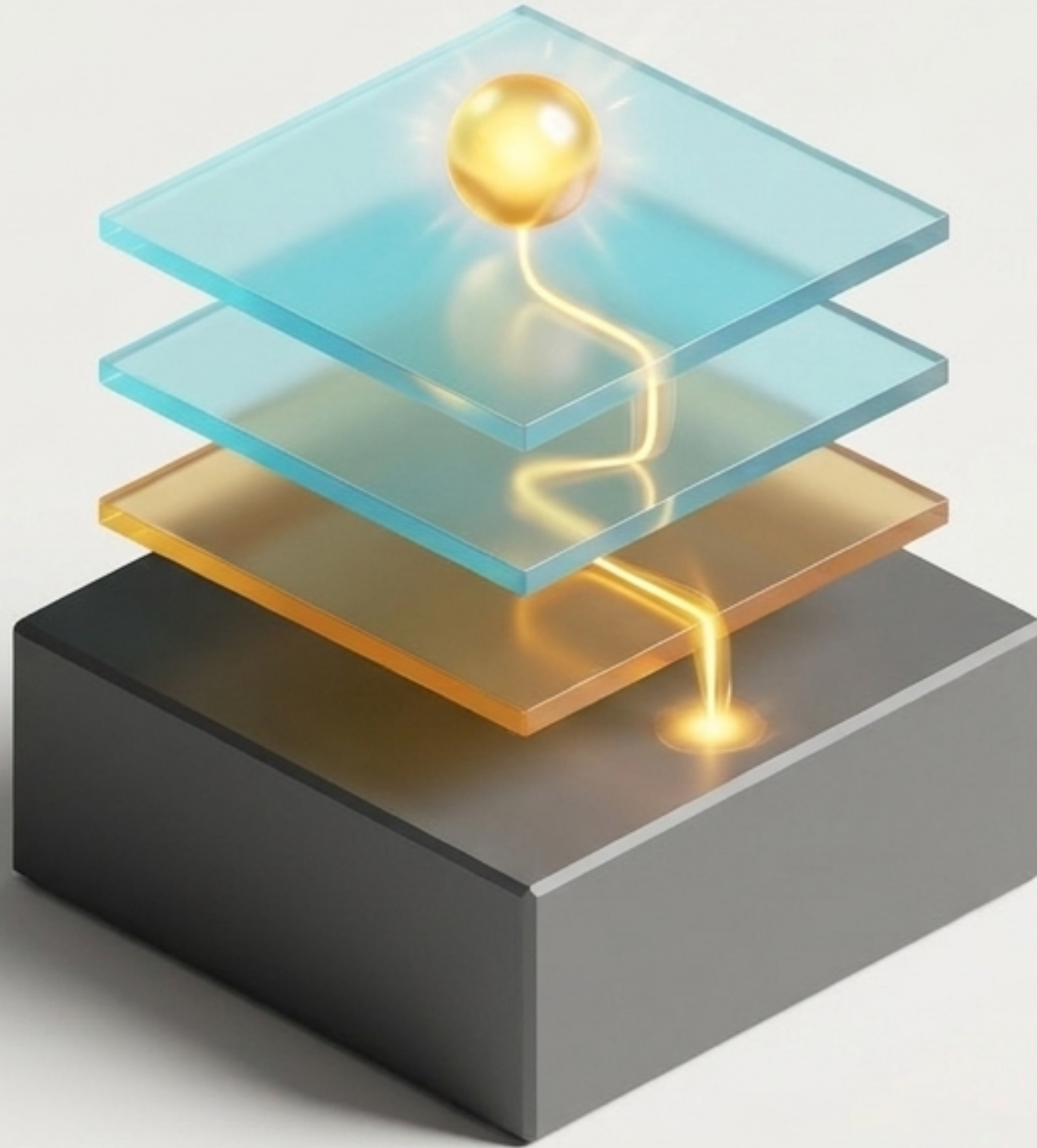
Ensure replicas sit on separate physical hardware, power supplies, and racks.



Test Recovery Constantly:

An architecture is only as valid as its last successful bare-metal restore.

The Complete Blueprint: Tracing the I/O Path



Layer 1 (Guest): Database App issues write request -> OS Filesystem processes it.



Layer 2 (Virtualization): VirtIO Controller translates it -> Hypervisor writes to the Thin Provisioned VMDK.



Layer 3 (Network): VMDK sits on an NFS Datastore -> Dedicated Storage Network routes the traffic, separated from VM traffic.



Layer 4 (Physical): Hardware Array receives it -> RAID 5 controller calculates parity and writes across multiple physical disks.

Mastery of virtualization is the ability to track the data across every unseen boundary.